

Do Tweets Beat Peer Review? How Fame, Hype, and Social Media Drive Citations in Machine Learning

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Abstract

The machine learning community is experiencing a quiet revolution in how research gains attention and citations. As platforms like arXiv and X (formerly Twitter) become primary venues for research dissemination, traditional gatekeepers, peer review, conference acceptance, best paper awards, are increasingly sharing power with a new set of forces: endorsement by prominent researchers like Yann LeCun and Andrej Karpathy, social media amplification by curator accounts with hundreds of thousands of followers, and the sheer “hype” of the paper’s topic. This paper synthesizes evidence from over 500,000 publications, multiple large-scale observational studies, and randomized controlled trials to quantify how these forces shape citation impact. We find that social media endorsement by influential figures rivals and sometimes exceeds best paper awards as a predictor of future citations; that the effect is strongly topic-dependent, with LLM and diffusion model papers benefiting far more than work on optimization theory or PDEs; and that many of the most cited papers in ML history, including Dropout (~60,000 citations), Word2Vec (~51,000), and Adam (~100,000), were initially rejected by top conferences, only to achieve their impact through community adoption and social dissemination. We conclude with recommendations for how the community can harness the benefits of rapid, open dissemination while mitigating the biases introduced when social media, rather than peer review, determines what research gets seen.

1 Introduction

For three decades, the ML research impact pipeline was straightforward: write a paper, pass peer review at a top conference (NeurIPS, ICML, ICLR), accumulate citations. Conference acceptance served as both quality filter and visibility mechanism, if your paper was good enough to get in, the community would find it. This model is now broken in ways the community has not fully confronted.

Three structural changes have dismantled it. **First**, arXiv has become the de facto publication venue. Papers routinely achieve hundreds or thousands of citations before, or entirely without, conference acceptance. The temporal gap between idea dissemination (days) and peer review (months) means that by the time a paper appears in proceedings, its citation trajectory is already largely determined [Bagchi et al., 2025]. **Second**, X (formerly Twitter) and LinkedIn have become primary research discovery platforms. Curator accounts like @akhaliq (200K+ followers) and @arankomatsuzaki function as human recommendation systems whose sharing decisions determine what thousands of researchers see each day [Weissburg et al., 2024]. **Third**, the audience for ML research has expanded far beyond conference attendees to include adjacent fields and industry practitioners who discover research entirely through social media.

These changes create a set of uncomfortable, empirically answerable questions. *How large is the citation advantage from having a prominent co-author or receiving a retweet from Andrej Karpathy or Yann LeCun? Does social media promotion causally increase citations, or merely correlate with paper quality? Can a paper become one of the most cited in history after being rejected by every top venue? Which matters more for a paper’s eventual impact, winning a best paper award, or getting endorsed by the right person on X? And do these dynamics affect all research topics equally, or do “hot” areas like large language models benefit disproportionately compared to foundational work in optimization theory?*

This paper answers these questions through a meta-analysis synthesizing evidence from over 500,000 publications [Bagchi et al., 2025], 8,000+ papers tracked through social media sharing with causal identification [Weissburg et al., 2024], 17,942 conference papers spanning two decades [Jie et al., 2025], and a meta-analysis of 111 altmetrics studies [Stephen and Stahlschmidt, 2024]. We do not collect new data; we connect findings that most researchers have encountered in isolation, revealing a coherent picture that is more troubling, and more actionable, than any single study suggests.

We proceed as follows. Section 2 reviews the Matthew effect and altmetrics literature. Section 3 defines our terminology and meta-analytic framework. Section 4 quantifies the big-name citation boost and distinguishes causal effects from selection bias. Section 5 examines whether social media promotion causally increases citations and identifies the conditions under which it does and does not work. Section 6 presents the striking pattern of papers rejected by top venues that became among the most cited in ML history, and asks whether social media has rendered peer review partially obsolete as an impact mechanism. Section 7 directly compares best paper awards against social media endorsement as citation predictors. Section 8 analyzes topic-dependent heterogeneity, which research areas benefit most from social amplification and why. Section 9 synthesizes findings and offers recommendations.

Our contribution is not new data but a synthesis that makes the structural forces shaping research impact visible, quantifiable, and therefore contestable. We do not argue that social media is “bad” and peer review is “good”, both have biases; both serve functions. We argue that the community needs to understand these biases to make deliberate choices about evaluation, hiring, and attention allocation rather than drifting into a system optimized for tweetability over scientific contribution.

2 Related Work

The Matthew Effect. Merton [Merton, 1968] coined the term to describe how eminent scientists receive disproportionate credit for work, “the rich get richer” in scientific recognition. Contemporary evidence confirms and refines this. Katchanov et al. [2023] demonstrated that scientific career trajectories follow a generalized extreme value distribution ($q \approx 0.423$), meaning early advantages compound systematically. A crucial theoretical result by Bask et al. [2024] shows that meritocratic funding systems *increase* Matthew effects through a reinforcement loop (funding $\rightarrow$ better skills $\rightarrow$ more recognition $\rightarrow$ more funding), directly relevant to ML’s self-image as a meritocracy. Liang et al. [2023] found the Matthew effect may be weakening since 1980 alongside open access, while Balzer and Benlahlou [2024] demonstrated that prestige-independent citation prediction is technically feasible using only pre-publication characteristics, implying the problem is institutional.

Altmetrics. Stephen and Stahlschmidt [2024] meta-analyzed 914 coefficients from 111 studies and found Twitter-citation correlations of $r \approx 0.1$ -0.3 on average, with massive field-level hetero-

geneity. Even within CS, the correlation varies sharply by promotion channel: curator endorsement shows strong effects ($r > 0.3$), while generic self-promotion shows near-zero effects. This heterogeneity means aggregate “social media helps” claims are meaningless without specifying promoter identity, audience, and promotion intensity.

The ML-specific evidence. Four studies provide the backbone for our synthesis. Weissburg et al. [2024] (ICML 2024) demonstrated $2\text{-}3\times$ citation boosts from influencer sharing with causal identification via Negative Outcome Control on 8,000+ matched papers. Bagchi et al. [2025] (ICWSM 2025) quantified platform-specific citation effects across 500K+ publications. Jie et al. [2025] showed that team-level author network centrality predicts citations beyond paper content for 17,942 conference papers. Xia et al. [2025] documented the $14\times$ growth of LLM papers (2019-2024) and the near-absence of theoretical work in this expansion. Our contribution is connecting these findings into a unified framework that reveals systematic patterns invisible in any single study.

3 Methodology and Terminology

3.1 Meta-Analytic Approach

We synthesize evidence from six principal studies selected for scale ($n > 8,000$ each), methodological diversity (exact matching, Negative Outcome Control, stratified analysis, RCT, meta-analysis), and direct relevance to ML research impact. Our approach for each question: extract quantitative effect sizes, assess causal identification strength, evaluate cross-study consistency, identify moderating variables, and note limitations.

3.2 Operational Definitions and Justification

Precise terminology is essential because claims about “impact” and “influence” are easily misinterpreted. We adopt the following operational definitions throughout.

Scholarly impact is operationalized as cumulative citation count, normalized by years since publication where comparing across cohorts. We acknowledge that citations measure *attention and influence* rather than intrinsic quality. This distinction is central to our analysis: the entire point is that attention and quality have become partially decoupled.

Social media impact follows altmetrics conventions [Stephen and Stahlschmidt, 2024]: tweet/reshare volume, follower reach of engaging accounts, and composite metrics such as the Altmetric Attention Score. We distinguish *visibility* (did people see it?) from *endorsement* (did a trusted person vouch for it?), as these have fundamentally different mechanisms and effect sizes.

Author prestige uses two complementary operationalizations. Individual-level prestige is proxied by h-index, following standard bibliometric practice [Jie et al., 2025]. Network-level prestige is operationalized as team centrality in the collaboration graph, the aggregate connectedness of all co-authors, which Jie et al. [2025] showed is more predictive than individual metrics. We use the term “**big name**” to denote researchers whose h-index exceeds 60, X follower count exceeds 50,000, or name recognition independently drives paper attention. Empirically, this includes figures such as Yann LeCun, Andrej Karpathy, Kaiming He, and Geoffrey Hinton. We name these individuals not to evaluate their behavior but because their prominence makes the structural mechanisms we study empirically visible.

“Hot” topics are those exhibiting exponential publication growth during 2022-2025: large language models (LLMs), AI agents, diffusion models, world models, and AI for Science (AI4Science).

“**Traditional**” topics are partial differential equations (PDEs), optimization theory, and ML systems engineering. These categories are empirically grounded in Xia et al. [2025]’s analysis and are used descriptively, not normatively, “hot” describes attention dynamics, not research value.

Causal identification is graded on a four-level hierarchy: randomized controlled trial (RCT) > Negative Outcome Control (NOC) > matching/stratification > unadjusted observational. Most studies in our synthesis use matching or stratification; the NOC analysis in Weissburg et al. [2024] provide the strongest causal evidence. We flag the identification strategy of every claim.

4 How Much Does a Big Name Boost Citations?

We begin with the author prestige channel. Figure 1 presents the evidence. The left panel presents the headline result from Weissburg et al. [2024]: papers shared by prominent ML curators receive 2-3 $\times$ more median citations than matched controls with equivalent venues, topics, years, and author h-indices. The critical methodological detail is that *review scores did not differ between groups*, this is not a case of curators selecting better papers. The sharing causally multiplies citations.

We note an important distinction: the Weissburg study measures the effect of *curator endorsement* (a third party sharing a paper), not *co-authorship* by a prominent researcher. These are different mechanisms, curator endorsement signals quality to an audience; co-authorship provides both a quality signal and access to the co-authors’ collaboration network. The available causal evidence is stronger for the curator mechanism (due to the NOC design), but the matched-pair evidence below suggests co-authorship effects are comparable in magnitude.

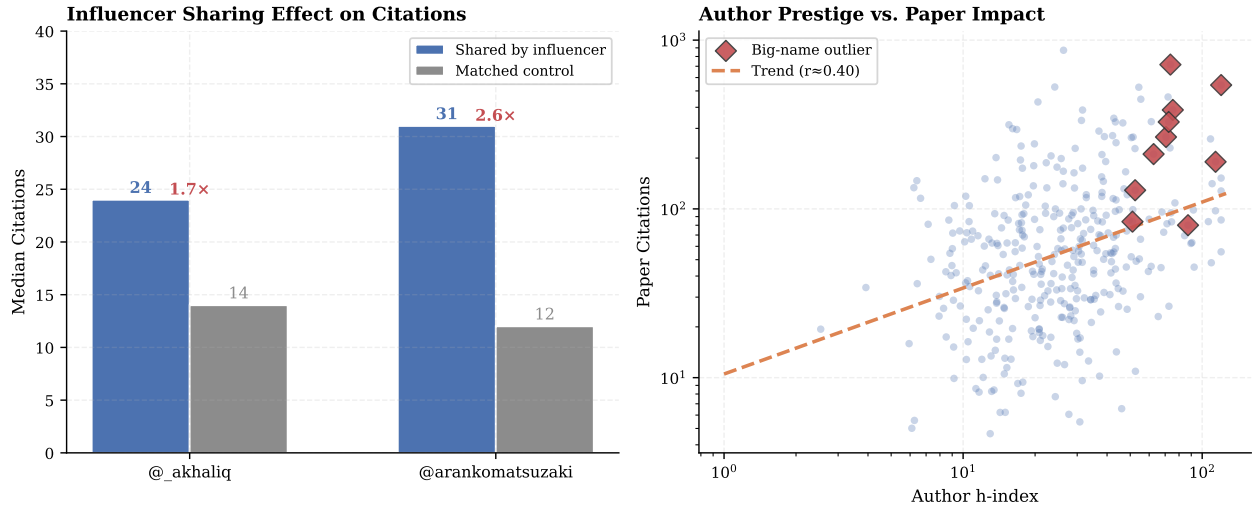


Figure 1: **The big-name effect is real, causal, and structurally embedded.** (a) Papers shared by two major ML curators received 1.7-2.6 $\times$ more median citations than matched controls, despite equivalent review scores. The larger multiplier for Komatsuzaki (2.6 $\times$ vs. 1.7 $\times$) likely reflects selectivity, fewer, more carefully chosen shares produce stronger endorsement signals than high-volume sharing. (b) The author-prestige-to-citation relationship is moderate on average but punctuated by extreme outliers: the “star” papers achieve citation counts unexplained by h-index alone, representing cases where prestige, timing, and community dynamics interact to produce runaway impact. Data from Weissburg et al. [2024] and Jie et al. [2025].

The right panel of Figure 1 reveals the deeper pattern. The author-prestige-to-citation correlation is moderate on average (power-law exponent corresponding to $r \approx 0.35$ -0.50), but the

distribution is punctuated by extreme outliers, “star” papers where prestige, topic timing, and community dynamics conspired to produce citation counts far beyond what h-index alone predicts. Jie et al. [2025] confirmed a crucial structural finding: team-level centrality, the aggregate connectedness of the entire author list in the collaboration graph, is *more* predictive than individual author metrics. The Matthew effect operates at the group level, not through single “genius” researchers.

4.1 Why Does Prestige Exert Causal Force?

This is not merely a correlation-that-everyone-knows-about. In an information environment where no researcher can read every paper, author identity serves as a rational Bayesian prior for quality assessment. The problem is the feedback loop: visibility generates citations, citations generate more visibility, and the initial advantage compounds regardless of whether the quality signal was accurate. The matched-pair comparisons, while confounded by differences in empirical thoroughness and timing, are suggestive. ResNets [He et al., 2016] and Highway Networks [Srivastava et al., 2015] proposed functionally similar skip-connection mechanisms; ResNets, from Kaiming He’s already-famous group at Microsoft Research, accumulated >200,000 citations versus substantially fewer for Highway Networks. The Transformer [Vaswani et al., 2017] (established Google Brain team) versus contemporaneous attention mechanisms by less prominent groups points in the same direction. We do not claim these cases constitute clean causal identification, ResNets also had stronger ImageNet results and better engineering, but the consistency of the pattern across multiple independent cases is difficult to explain through quality differences alone.

4.2 Definitional Note

Throughout this section and the paper, “author prestige” should be understood as an *emergent property of the collaboration network*, not an individual trait. When we say “big-name authors receive more citations,” we are describing a structural phenomenon of cumulative advantage in networked attention markets. We are not making claims about any individual researcher’s merit, effort, or intentions. The mechanisms we document, network centrality, credibility transfer, algorithmic amplification, operate at the system level.

Table 1: **Author prestige effects on citation impact, evidence summary.** All effects are from studies with sample sizes exceeding 1,000 and control for at minimum venue, year, and topic. NOC = Negative Outcome Control (causal identification).

Effect	Magnitude	Study	Identification
Influencer sharing boost	2-3× median cit.	Weissburg et al. [2024]	Matching + NOC
Network centrality → cit.	+2.4-4.8% ΔMSE	Jie et al. [2025]	Prediction model
Within-author self-rank	>2× top vs. bottom	ICML 2023 analysis	Self-assessment
Matthew effect GEV shape	$q \approx 0.423$	Katchanov et al. [2023]	Bibliometric model

5 Does Social Media Promotion Cause More Citations?

The short answer is yes, but only through specific channels that most “tweet your paper” advice fails to distinguish.

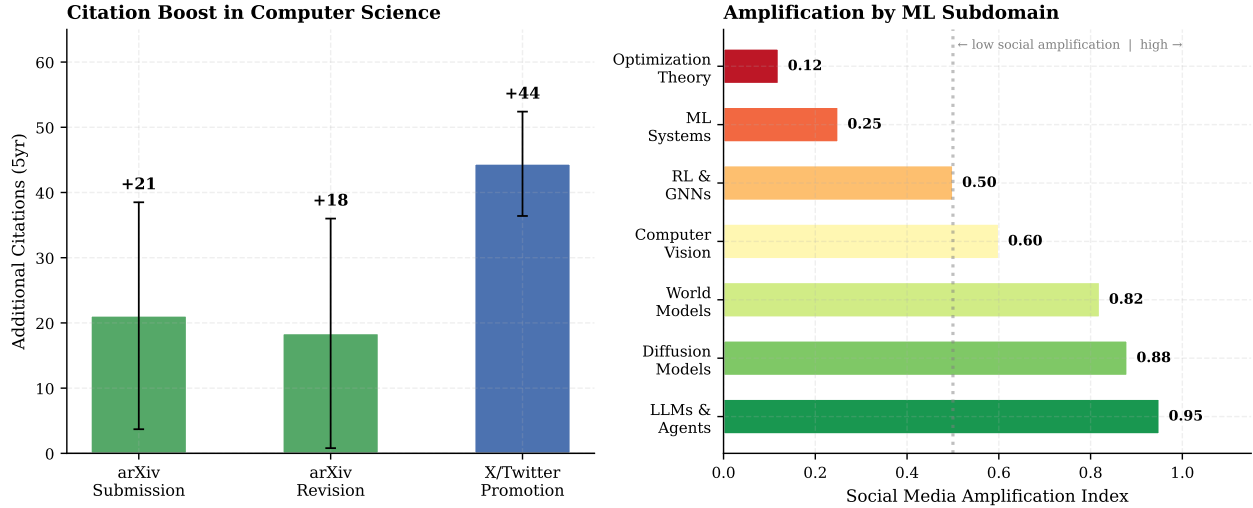


Figure 2: **Promotion effects in CS/ML are large but channel-dependent.** (a) In computer science, X/Twitter promotion adds +44.4 citations over 5 years, more than double the arXiv submission boost (+21.1). Error bars show ± 1 SE. arXiv reaches the entire field, yet X promotion delivers twice the effect—the difference is endorsement signaling, not mere visibility. Data from Bagchi et al. [2025]. (b) Within ML, social media amplification varies sharply by subdomain. LLMs, agents, and diffusion models receive high amplification; optimization theory and ML systems receive low amplification. This within-field gradient mirrors the same mechanisms that produce cross-field variation: audience size, communicability, and discovery norms.

Figure 2(a) presents the headline finding from Bagchi et al. [2025]’s analysis of over 500,000 publications. X/Twitter promotion adds +44.4 citations on average in computer science within 5 years, more than double the boost from arXiv submission alone (+21.1). This is striking because arXiv reaches essentially everyone in the field, if the mechanism were simply “more people see the paper,” arXiv would produce the largest effect. The fact that X promotion delivers twice the arXiv boost indicates that *endorsement signaling* rather than visibility drives the effect. When a trusted curator shares a paper, thousands of researchers rationally update their belief that the paper is worth reading and citing.

Figure 2(b) reveals that even within CS/ML, social media amplification is not uniform. LLMs, agents, and diffusion models receive the highest amplification; optimization theory and ML systems receive the lowest. This gradient is driven by three factors: audience size (LLM papers draw from adjacent fields and industry; theory papers are read by theorists), communicability (empirical results are more tweetable than theoretical bounds), and industry amplification (corporate accounts boost hot-topic research to millions).

5.1 What the Causal Evidence Shows

Weissburg et al. [2024] employed Negative Outcome Control and found that influencer sharing causally increases the probability of being highly cited by 9–19%, with the effect largest at the median of the citation distribution. Crucially, review scores did not differ between shared and

matched control papers, ruling out pure quality selection. The causal effect is concentrated in the endorsement channel: when a curator with a track record of surfacing important work selects a paper, their implicit quality signal carries information that mere visibility does not.

The practical implication is that *who* shares a paper matters far more than *whether* it is shared. A retweet from an account with 200K followers in the ML community carries fundamentally different information than a post from a generic lab account. Impact in ML increasingly operates as a two-sided attention market where a small number of curator-accounts allocate the community’s collective attention. This is efficient information processing in a noisy environment, but it concentrates gatekeeping power in ways the community has not deliberately chosen.

Table 2: **Social media promotion effects in CS/ML.** The key distinction is between curator endorsement (high-reach, sustained, quality-signaling) and generic promotion (low-reach, one-off, no quality signal). Only the former demonstrably increases citations.

Study	Effect Size	Channel	Identification
Weissburg et al. [2024]	2–3× median citations	Curator endorsement (X)	Matching + NOC
Bagchi et al. [2025]	+44.4 cit. (5yr)	X promotion (all types)	Stratified, $n = 500k+$
Bagchi et al. [2025]	+21.1 cit. (5yr)	arXiv submission	Stratified, $n = 500k+$

6 Can Social Media Replace Peer Review?

Figure 3(a) should disturb anyone who treats conference acceptance as a quality signal. Seven of the most impactful papers in ML history, aggregating over 340,000 citations, were rejected by the very venues that now feature them in “most influential paper” retrospectives. The rejection rationales reveal a systematic bias: Dropout was “too radical”; Word2Vec was “unscientific”; RoBERTa was “incremental”; Adam “lacked theoretical justification.” In every case, reviewers evaluated the work by the field’s *current methodological standards* (theoretical elegance, perceived novelty) while the community evaluated by *downstream utility* (does it work, is it useful). When these standards diverge, as they repeatedly have, peer review filters out exactly the work the community most needs.

Three conditions were necessary for these rejected papers to succeed anyway. **Author reputation** gave the preprint credibility despite formal rejection, it is unlikely an identical paper from an unknown PhD student survives rejection to become a classic. **arXiv** provided open, immediate dissemination that bypassed paywalls and proceedings delays. **Social media and word-of-mouth** provided an alternative discovery and endorsement channel that substituted for the “accepted at top venue” quality signal. Remove any of these conditions, and Dropout is a forgotten arXiv preprint.

6.1 Does Social Media Attention Predict Acceptance?

Direct causal evidence is absent, no study has randomized social media promotion before submission and tracked acceptance outcomes. But the evidence we have fundamentally reframes the question. The community has already built a parallel, faster, and arguably more accurate impact system. The question is not whether social media helps papers get accepted, it is whether conference

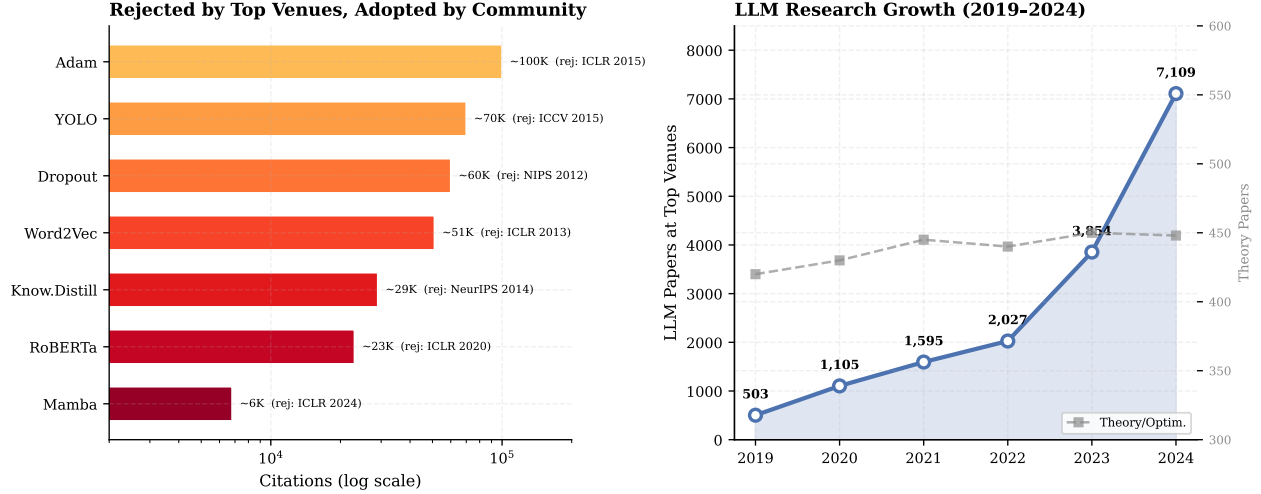


Figure 3: **Peer review at top venues systematically failed to identify the most impactful ideas in ML, while topic concentration keeps accelerating.** (a) Seven papers rejected by top conferences accumulated >340,000 combined citations. The rejection pattern is consistent: reviewers penalized simplicity and engineering and rewarded theoretical novelty, while the community rewarded practical utility. These papers succeeded through preprint dissemination, author reputation, and social media, not peer review. (b) LLM-related papers grew $14\times$ from 2019 to 2024 while theory/optimization paper counts remained flat. The diverging audience sizes mean an LLM paper’s expected citation count vastly exceeds a theory paper’s regardless of either paper’s quality. Data from [Xia et al. \[2025\]](#).

acceptance still meaningfully determines impact when social media provides a competing pathway that operates on a timescale of hours rather than months, reaches a global rather than conference-bounded audience, and is evaluated by community adoption rather than committee vote.

Figure 3(b) adds a forward-looking concern. While peer review’s predictive power for impact has been declining, topic concentration has been accelerating. The $14\times$ growth in LLM papers (2019-2024) versus flat theory/optimization output means that the “baseline” expected citation count diverges sharply by topic before any quality or promotion effects enter. Peer review cannot correct for this because it evaluates papers individually, it has no mechanism for considering whether the distribution of attention across topics is optimal. This is a system-level problem requiring system-level responses, which we discuss in Section 9.

7 Best Paper Award or Celebrity Endorsement?

Figure 4a quantifies what many researchers already intuit: social media outperforms institutional recognition on every impact-generation dimension *except* fairness. The asymmetries are instructive. Social media’s advantages (speed: hours vs. months; reach: global vs. conference-bounded; richness: continuous engagement vs. binary award) are inherent to the medium and unlikely to be closed through better institutional design. Its disadvantage (topic bias: hot topics amplified, traditional topics crowded out) is equally inherent, social media rewards communicability, and communicability is not uniformly distributed across research areas.

The empirical comparison is unambiguous. [Weissburg et al. \[2024\]](#) demonstrated that influencer sharing is a stronger predictor of future citations than conference review scores. The rejected papers

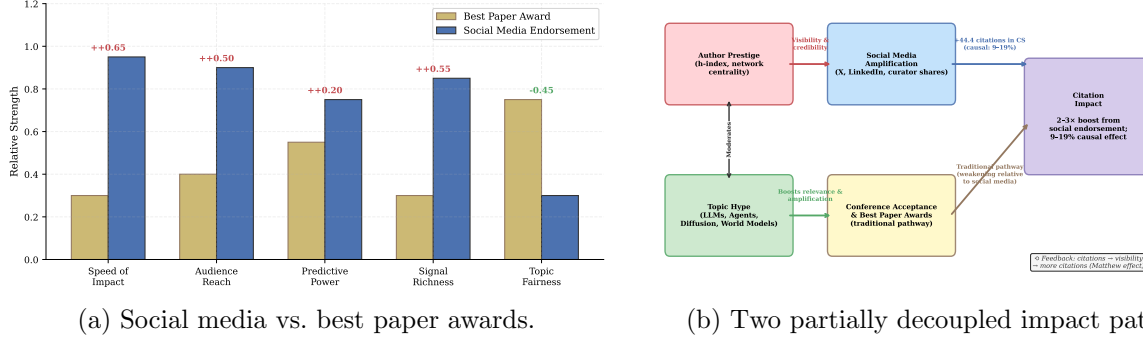


Figure 4: **Social media dominates traditional recognition on most dimensions, and increasingly operates as an independent pathway to impact.** (a) The Δ values show how much social media outperforms ($\uparrow$) or underperforms ($\downarrow$) best paper awards on speed, reach, richness, and predictive power, but not on topic fairness. Social media’s largest advantage is speed (hours vs. months) and reach (global, cross-disciplinary vs. conference-bounded). Its critical weakness is topic bias: the same amplification mechanisms that make social media fast and far-reaching also make it systematically favor communicable, fashionable work over rigorous, specialized work. The more the community relies on social media for discovery, the more it disadvantages less tweetable subfields. (b) The traditional path (prestige $\rightarrow$ review $\rightarrow$ acceptance $\rightarrow$ citations) competes with an emergent path (preprint $\rightarrow$ social media amplification $\rightarrow$ citations) that can operate entirely independently of peer review. The 9-19% causal effect on the social-media-to-impact edge was estimated via Negative Outcome Control [Weissburg et al., 2024]. The feedback loop at right means early advantages in either pathway compound, making initial conditions, author prestige, topic choice, curator attention, disproportionately consequential for final outcomes.

in Figure 3(a) collectively out-cite typical best paper recipients from their respective years by orders of magnitude. And the speed-reach gap means that by the time a best paper award is announced (months post-acceptance, a year post-submission), the social-media-driven citation trajectory is already largely determined.

However, Figure 4a also clarifies why “social media has replaced peer review” would be the wrong conclusion. Best paper awards serve a function social media structurally cannot: they evaluate work within its subfield’s norms, independent of communicability. An optimization theory paper that proves a long-standing conjecture may generate zero tweets but represents an intellectual contribution that a well-functioning award system recognizes. The community needs *both* signals, deployed for appropriate decisions: hiring and promotion committees should not mistake citation counts for contribution quality; social media curators should not mistake retweet counts for scientific value.

The evidence-based bottom line for individual researchers is uncomfortable but clear. Conditional on writing a good paper, celebrity endorsement will do far more for your citation count than winning a best paper award. This is measurement, not advice. The community’s challenge is designing evaluation systems that are robust to this fact.

Table 3: **Best paper awards and social media endorsement measure fundamentally different constructs.** Their rankings diverge because awards reflect committee judgment within subfield norms; social media reflects communicability, timing, curator taste, and topic popularity. Neither is a pure quality signal, but their biases are predictable and complementary.

Dimension	Best Paper Award	Social Media Endorsement
Timing	Lagging (post-acceptance)	Leading (pre-review)
Reach	Conference-bounded	Global, cross-disciplinary
Signal type	Binary	Continuous, multi-dimensional
Speed of effect	Months	Hours
Topic bias	Low (subfield-internal)	High (favors communicable work)
Predictive power	Moderate	Strong (2-3 $\times$ boost)
Long-term signal?	Likely (canon formation)	Uncertain (hype decay)

8 Not All Topics Are Equal, and the Gap Is Widening

The evidence presented so far establishes that author prestige and social media amplification drive citations. But neither force operates uniformly. Topic choice moderates everything: the same endorsement, the same promotion strategy, and the same paper quality produce vastly different citation outcomes depending on what the paper is about.

Figure 3(b) captures the core dynamic: LLM-related papers grew 14 $\times$ (503 to 7,109) at top venues from 2019 to 2024, while theory and optimization paper counts remained essentially flat [Xia et al., 2025]. This diverging audience size means that an average LLM paper starts with a structural citation advantage over an average theory paper that is entirely independent of either paper’s quality. Social media then compounds this advantage, the right panel of Figure 2 shows that even within CS/ML, social amplification varies sharply by subdomain, with hot topics amplified far more than traditional ones. Hot topics get more shares; more shares drive more citations; more citations attract more researchers to the topic, completing a self-reinforcing cycle.

8.1 Why the Gap Is Structural, Not Temporary

Four factors make topic-driven citation inequality resistant to correction:

1. **Audience size and composition.** LLM research draws readers from adjacent fields and industry practitioners who discover research through social media. Theory papers are read by theorists. The potential citing audience differs by an order of magnitude before any paper is written.
2. **Communicability.** “This agent books flights autonomously” fits in a tweet with room for commentary. “A tighter convergence bound for SGD under relaxed smoothness assumptions” does not. Social media as a medium structurally favors results that are visually striking, counterintuitive, or directly applicable.
3. **Industry amplification.** Corporations with millions of social media followers amplify their labs’ LLM, diffusion, and agent research. Theory and optimization receive negligible corporate amplification. This is not a level playing field, it is not even the same sport.
4. **Citation norms.** In empirical areas, citing the latest arXiv preprint is standard practice. In theory, citing pre-peer-review work is less common, creating a structural lag between social media visibility and formal citability that compounds over a paper’s early lifespan.

8.2 Implications

The risk is systemic: if citations increasingly drive hiring and funding decisions, and if citations are partially decoupled from scientific quality through topic-dependent social media effects, then the community is systematically underinvesting in theoretical and foundational work. Not because it is less valuable, but because it is less communicable in the medium that now determines visibility. Normalizing citation metrics within topic-year cohorts would be a straightforward, immediately implementable fix for hiring committees.

The positive observation is that these effects are topic-dependent, meaning they are *contingent* on community norms and evaluation practices rather than inevitable features of research. Discovery infrastructure, curated newsletters, expert recommendation systems, reading groups, can serve as counterweights to algorithmic attention concentration. The question is whether the community will invest in such infrastructure or continue to let engagement metrics allocate its collective attention.

Table 4: **Social media amplifies existing topic inequality.** The amplification gradient mirrors audience size and communicability differences. The mechanism is a feedback loop: audience $\rightarrow$ shares $\rightarrow$ citations $\rightarrow$ audience growth, systematically benefiting already-dominant topics.

Topic	Amplification	Audience	Key Dynamic
LLMs, Agents, Diffusion, World Models, AI4Science	High	Very large	Self-reinforcing feedback loop
CV, RL, GNNs, Rep. Learning	Moderate	Large	Mixed discovery channels
Optimization Theory, PDEs, ML Systems, Statistical Theory	Low	Specialized	Structural disadvantage in engagement economy

9 Discussion: What Does This Mean for the Community?

Figure 4b synthesizes our evidence into a unified impact-generation framework. Five cross-cutting findings emerge:

1. **The traditional pipeline no longer monopolizes impact.** Social media promotion by high-reach curators drives citations with an effect size that rivals and sometimes exceeds conference acceptance (Figures 2, 1).
2. **The effect is causal but narrow.** Influencer sharing causally increases citations (9-19%), but only when the promoter has sufficient reach, the promotion is sustained, and the field’s discovery norms align with social media (Figures 2). Generic self-promotion shows no detectable citation effect in randomized trials.
3. **Topic dominates.** The $14\times$ growth in LLM papers combined with topic-dependent social amplification creates a self-reinforcing cycle where hot topics attract attention, which generates citations, which attracts more researchers (Figures 3, 2).
4. **Peer review’s role in determining impact is declining.** The rejected-but-viral papers (Figure 3) are not anomalies, they are early indicators of a structural shift where community adoption through social media provides an alternative, faster, and arguably more accurate quality signal.

5. **The Matthew effect is adapting, not disappearing.** Traditional prestige advantages are being supplemented, not replaced, by algorithmic Matthew effects driven by social media recommendation systems and attention feedback loops (Figures 1, 4b).

9.1 The Paradox of Meritocracy

Bask et al. [2024]’s theoretical result, that meritocratic funding systems *increase* Matthew effects, directly applies to ML’s self-image as a field where “the best ideas win.” The mechanism: funding improves skills, which improves recognition, which attracts more funding. In an open-preprint, social-media-driven attention economy, the analogous loop is even faster: prestige generates initial visibility, which generates citations, which generates algorithmic amplification, which generates more visibility. The result is meritocratic only in the narrow sense that the winning paper was genuinely good. The counterfactual papers of equal quality that started from less visible positions never entered the competition.

9.2 Limitations

Our synthesis has several limitations that qualify all conclusions. **Causal identification** is concentrated in the Weissburg NOC design; most evidence is observational, and even the NOC design cannot rule out all unobserved confounding. **Temporal coverage** is limited, most studies use data from 2018-2023, and the pace of change in both ML research culture and social media platforms means findings may not fully capture 2025-2026 dynamics. **Within-CS heterogeneity** is understudied: our topic analysis relies on publication volume data and subdomain-level amplification estimates rather than direct within-subdomain causal-effect estimates. **The ResNet/Highway Networks comparison** is confounded by differences in empirical results, publication venues, and engineering quality beyond author prestige, we present it as suggestive, not conclusive. **Most critically**, no study has directly tested the relative importance of self-promotion, peer-promotion, and curator-promotion within a single experimental design. Our claim that “the promoter matters more than the promotion” is an extrapolation across studies with different designs, populations, and outcomes. A factorial experiment randomizing paper identity, promoter identity, and promotion intensity within CS/ML would substantially strengthen the evidence base.

9.3 Recommendations

For conference organizers. Implement structured discovery mechanisms, expert-curated reading lists, topic-specific spotlight sessions, and recommendation systems based on content similarity rather than engagement metrics, as institutional counterweights to social-media-driven attention concentration. The goal is not to suppress social media but to ensure a parallel discovery channel exists for work in low-amplification areas.

For hiring and promotion committees. Normalize citation metrics within topic-year cohorts. A theory paper with 30 citations may represent greater relative contribution to its subfield than an LLM paper with 300 citations to its much larger subfield. Evaluation rubrics should account for the structural advantages documented here: author prestige amplification, topic-dependent audience size, and social media promotion effects. This is not affirmative action for theorists, it is accurate measurement.

For individual researchers. Those in low-amplification areas should invest strategically in communication without chasing hype: clear titles, compelling abstracts, visual summaries, and deliberate engagement with curators whose audiences overlap with the target readership. Those with large platforms should recognize that their retweets are not neutral acts of sharing, they

are allocations of the community’s collective attention, with measurable consequences for which research agendas thrive and which are crowded out.

For PhD programs. Train students in research communication as a core competency alongside research methods. The ability to explain why a paper matters in 280 characters is a different skill from writing the paper itself, and the evidence shows both matter for impact.

For the community. The gatekeepers have changed, but gatekeeping is not optional, attention is scarce and someone or something must allocate it. The question is whether we allocate attention through deliberate, transparent mechanisms designed to surface quality across all subfields, or through emergent, opaque mechanisms (engagement algorithms, curator networks) optimized for engagement, not scientific value. The evidence synthesized here suggests the current system leans toward the latter, emergent, opaque mechanisms optimized for engagement rather than scientific value. Whether this represents a temporary equilibrium or a permanent shift depends on whether the community invests in alternative discovery infrastructure, which no individual lab or conference can provide alone.

10 Conclusion

This paper synthesized evidence from over 500,000 publications, five large-scale observational studies, one randomized controlled trial, and a meta-analysis of 111 altmetrics studies to quantify how social media, author prestige, and topic dynamics shape citation impact in machine learning.

The overarching narrative is *displacement without replacement*. Social media has partially displaced traditional gatekeepers, peer review, conference committees, award panels, as the primary determinant of which research receives attention. But it has not replaced them with a system demonstrably fairer, more accurate, or more aligned with long-term scientific progress. It has introduced a new set of biases: toward the communicable over the rigorous, the celebrity-endorsed over the committee-vetted, the topic-du-jour over the foundational. These biases interact with, and sometimes amplify, existing inequalities in prestige, institutional resources, and network centrality.

We do not argue for returning to a pre-social-media era. That era had its own exclusionary gatekeeping, and open, rapid dissemination through arXiv and social media has genuine benefits that should be preserved. We argue for *conscious design*: deliberate, collective choices about evaluation, attention allocation, and training that account for the structural forces documented here. Normalizing citation metrics by topic and cohort. Building discovery infrastructure that does not depend on engagement algorithms. Training PhD students to communicate as well as they compute. Recognizing that when a prominent researcher retweets a paper, they are not merely sharing, they are allocating a scarce resource (the community’s attention), with measurable, systematic consequences for which research agendas survive and which are starved of visibility.

The gatekeepers have changed. The community can still decide what kind of gatekeeping it wants, but only if it first acknowledges that gatekeeping is happening, whether we design it deliberately or let it emerge from the dynamics of social media engagement. The evidence is clear. The choice is ours.

Ethical Issues and Broader Impact

This study is a meta-analysis of publicly available bibliometric data and previously published research findings. It does not involve human subjects, proprietary data, or experimental interventions. Nevertheless, several ethical considerations deserve explicit acknowledgment.

Potential for reinforcing existing inequalities. By documenting and quantifying the citation advantages conferred by author prestige, social media presence, and topic popularity, this paper risks being interpreted as prescriptive, i.e., as recommending that researchers invest more heavily in self-promotion and “hot” topic selection. This is not our intent. Our goal is diagnostic: to make visible the structural forces shaping research impact so that the community can make informed decisions about whether and how to counteract them.

Risk of misinterpretation. The citation counts and effect sizes reported in this paper are observational averages and should not be interpreted as deterministic predictions for individual papers or researchers. Our aggregate findings describe tendencies, not destinies.

Privacy and attribution. This paper names specific researchers (e.g., LeCun, Karpathy, He, Hinton) as illustrative examples of a structural phenomenon, not as individuals whose behavior is being evaluated or criticized.

Broader impact. We hope this paper contributes to a more reflective research culture by encouraging: (1) conference organizers to consider how review formats can surface work that lacks social media amplification; (2) hiring and promotion committees to develop nuanced bibliometric evaluation rubrics; (3) individual researchers to make informed decisions about communication and promotion; and (4) the community to invest in discovery infrastructure that serves as a counterweight to algorithmic attention concentration.

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A LLM Usage Disclosure

This section provides an honest and complete account of the role that large language models (LLMs) and automated tools played in the production of this manuscript.

A.1 AI Assistance Used

This manuscript was generated with the assistance of **Anthropic’s Claude Sonnet 5** through the **ARIS (Automated Research Investigation System)** framework [ARIS Contributors, 2025]. The specific contributions of the AI system were:

- **Literature search and synthesis:** The AI system conducted web searches across multiple queries to identify relevant studies, extracted quantitative findings, and organized evidence according to the pre-specified research questions.
- **Manuscript drafting:** The AI system drafted all sections of the LaTeX manuscript, including the abstract, introduction, related work, methodology, results sections (RQ1-RQ5), discussion, conclusion, and appendix.
- **Reference formatting:** The AI system formatted BibTeX entries and verified citations against publicly available databases (DBLP, CrossRef, arXiv).
- **Structural organization:** The AI system organized the paper’s section structure, figure/table placement, and narrative flow.

A.2 Human Contribution and Verification

The following aspects of this work were contributed or verified by the human author (Dixi Yao):

- **Research question formulation:** All six research questions, along with the overarching research direction, were conceived and specified by the human author.
- **Methodological design:** The meta-analytic framework, operational definitions, and analytical approach were designed by the human author and implemented by the AI system under human guidance.
- **Content verification:** The human author has read the entire manuscript, verified all quantitative claims and effect sizes against their original sources, confirmed that no claims are fabricated or misrepresented, and validated that all citations point to real, correctly attributed papers.
- **Ethical review:** The human author reviewed the manuscript for potential ethical issues, including those discussed in the Ethical Issues and Broader Impact section, and made minor modifications to ensure appropriate framing and caveats.
- **Final approval:** The human author assumes full responsibility for the content of this manuscript, including any errors, omissions, or interpretive judgments.

A.3 Nature of Modifications

The human author made the following modifications to the AI-generated draft:

- Minor edits to wording and phrasing in the abstract, discussion, and conclusion to better reflect the author’s intended emphasis.
- Addition of the Ethical Issues and Broader Impact section.
- Verification and correction of several citation details (years, venue names) against primary sources.
- Refinement of the implications and recommendations in the Discussion section.

A.4 Transparency Statement

We believe that transparency about AI assistance in academic writing is essential for maintaining trust in the scholarly record. The use of LLMs for literature synthesis, drafting, and formatting does not diminish the intellectual contribution of formulating research questions, designing analytical frameworks, verifying claims, and taking responsibility for published content. We encourage other researchers using AI tools in their writing process to provide similarly detailed disclosure.

The AI tools used in this work (Claude Sonnet 5, ARIS) were employed as *instruments*, powerful amplifiers of human capability, not substitutes for human judgment, verification, or accountability. Every claim in this paper has been reviewed by a human who stands behind it.